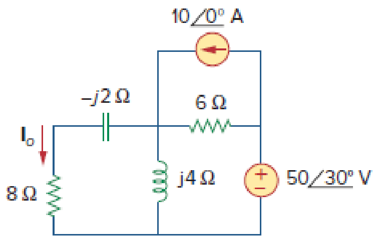
Chapter 10, Problem 3PP

Problem

Find **I***o* in Fig. 10.8 using mesh analysis.

Figure 10.8:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.8 in the textbook.

Consider the following circuit diagram:

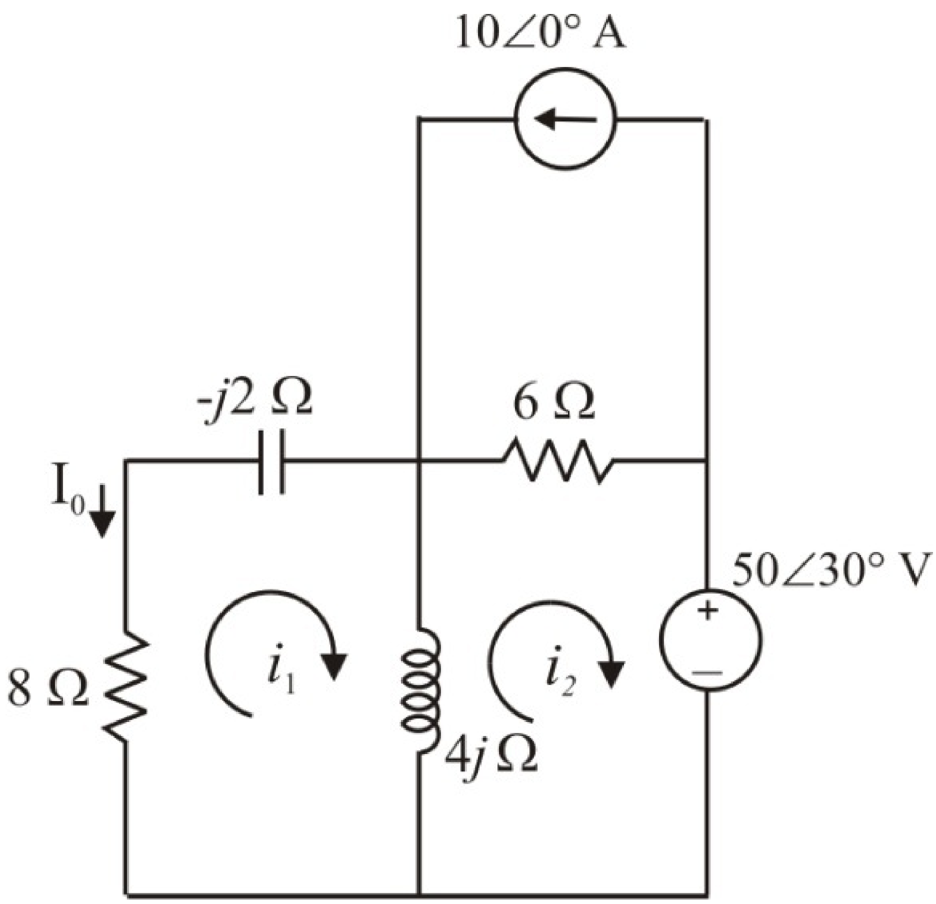


Figure 1

Step 2 of 4

Assume that the currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are flowing in the circuit of Figure 1.

Apply Kirchhoff's voltage law to mesh 1.

$$(8\ \Omega)\mathbf{I}_1 - (j2\ \Omega)\mathbf{I}_1 + (j4\ \Omega)(\mathbf{I}_1 - \mathbf{I}_2) = 0$$

$$\mathbf{I}_1(4+j) - j2\mathbf{I}_2 = 0 \ \ldots\ldots (1)$$

Apply Kirchhoff's voltage law to mesh 2.

$$(j4\ \Omega)(\mathbf{I}_2 - \mathbf{I}_1) + (6\ \Omega)(10 + \mathbf{I}_2) + 50\angle 30° = 0$$
$$-j4\mathbf{I}_1 + (6 + j4)\mathbf{I}_2 + 60 + 50\angle 30° = 0$$

$$j4\mathbf{I}_1 - (6 + j4)\mathbf{I}_2 = 103.02 + j24.92 \ \ldots\ldots (2)$$

Step 3 of 4

Write the equations (1) and (2) in matrix form.

$$\begin{bmatrix} 4+j & -j2 \\ j4 & -6-j4 \end{bmatrix}\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 103.02 + j24.92 \end{bmatrix}$$

Calculate the determinant values of matrices to find the values of $\mathbf{I}_1$ and $\mathbf{I}_2$.

$$\begin{aligned}\Delta &= \begin{vmatrix} 4+j & -j2 \\ j4 & -6-j4 \end{vmatrix} \\ &= (4+j)(-6-j4) - (-j2)(j4) \\ &= -(24 + j16 + j6 - 4) - 8 \\ &= -28 - j22\end{aligned}$$

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} 0 & -j2 \\ 103.02 + j24.92 & -6-j4 \end{vmatrix} \\ &= 0(-6-j4) - (-j2)(103.02 + j24.92) \\ &= -49.85 + j206.05\end{aligned}$$

Step 4 of 4

Calculate the current $\mathbf{I}_o$ in mesh 1 of circuit in Figure 1:

$$\begin{aligned}\mathbf{I}_o &= -\mathbf{I}_1 \\ &= \frac{-\Delta_1}{\Delta} \\ &= \frac{-(-49.85 + j206.05)}{-28 - j22} \\ &= 5.95\angle 65.44°\ \text{A}\end{aligned}$$

Therefore, the value of current $\mathbf{I}_o$ is $\boxed{5.95\angle 65.44°\ \text{A}}$.